

Problem Set 1

Due date: September 23rd, 17:00.

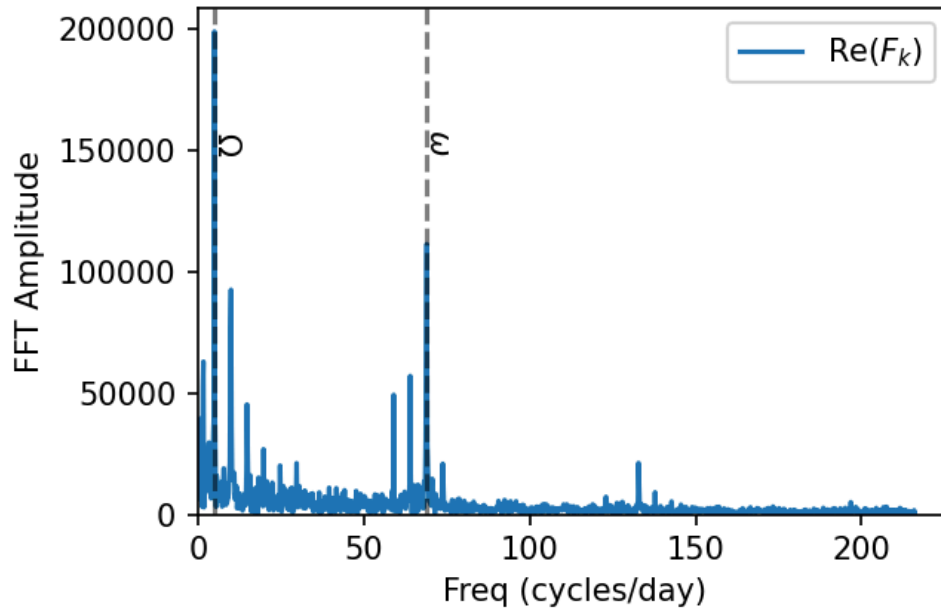
Please upload all answers to Canvas. This includes pdf scans of whatever hand written solutions you produce, any code which you write (including instructions on how to execute that code) and any outputs that code produces. For all of your code please be sure it is fully commented, and for any plots be sure that axes have labels and are legible.

Problem 1 - The Fourier Transform

1. Write a piece of code which calculates the discrete Fourier transform. Verify your code works by calculating the Fourier Transform of $y(t) = \cos(2\pi 2t)$. You are free to define your own internals and time resolution for t . The output of this program should be a plot of $y(t)$ versus t and a plot of $Y(f)$ versus f . The plot of the Fourier spectrum should also have the known periods marked with vertical lines. (4 marks)
2. Compute the DFT for $y(t) = \cos(2\pi 2t) + \sin(2\pi 4t)$ for the following intervals (min,max,step):
 1. $t = \text{range}(0,4,0.5)$
 2. $t = \text{range}(0,4,0.1)$
 3. $t = \text{range}(0,20,0.1)$

Using these results, comment on the accuracy of the recovered periods as both a function of the number of samples and the time step between the samples. (5 marks)

3. Distributed with this problem set is a time series of a binary star system. Verify whether the condition to apply a FT to these data are met. If it is not, write a piece of code which can manipulate the data to meet this condition. (2 marks)
4. The Fourier Transform can be used as a high pass or low pass filter to isolate signals of interest. The time series discussed in the previous problem displays multiple periodicities. These can be broken down into two main components: a series of low frequency components ($f < 20$ per day) due to the orbital motion of the companion star in the binary, and some high frequency components ($f \sim 68$ cycles per day). By using appropriate frequencies filters, produce a plot of each of the main signals versus time. You can use a FFT package if necessary (np.fft in python or fftw3.h in C++). (5 marks)
5. The two main periods in the frequency spectrum of the previous example are at a frequency of ~ 5 cycles per day (which we will label as Ω) and ~ 69 cycles per day (which we will label as ω). Identify the relationships between all other significant peaks and these two periods in the spectrum and submit a plot with all of these peaks clearly labelled (an example is given below with Ω and ω labelled). (2 marks)



Problem 2 - A Monte Carlo Radiative Transfer code

1. In class, you were shown that the relation between ξ and μ/ϕ was

$$\mu_0 = 2\xi - 1$$

$$\phi_0 = 2\pi\xi$$

Show that this is true (submit your answer as a scan of a pen and paper solution). (2 marks)

2. Write a Monte Carlo code which simulates isotropic scattering of photons through a uniform slab of thickness $z_{\max} = 1$ and $\tau_{\max} = 10$. The photons should all be initialised at the origin with an initial angle of direction aligned with the z axis. They should then move a distance which is related to a randomly drawn optical depth, and then scatter into a random direction. The ending criterion for considering a photon should be when it leaves the atmosphere at $z = z_{\max}$ or back scatters out of the atmosphere. The final output of this program should be a histogram of the angles over which photons are emitted. Use this code to reproduce Chandrasekhar's result from 1960 for a plane atmosphere as shown below (and attached to this assignment as a dat file with columns of angle and intensity). Simulating roughly 100,000 photons should be enough to reproduce the result (depending on your bin size). Note that you'll need to convert from number of photons (flux) in a given bin to intensity. This can be done using

$$\frac{I_\nu}{F} = \frac{N_i N_\mu}{2N_0 \mu_i}$$

where N_i is the number of photons in bin i , N_μ is the total number of μ bins you split the data in to, N_0 is the total number of photons, and μ_i is the μ at the centre of the i th bin.

(10 marks)

3. Determine appropriate errors for each of your bins, and update your plot accordingly. Justify your choice of errors. (4 marks)

